Bumblebee Path Integration

What we are doing

We are testing whether bumblebees can orient under low-light conditions using polarised cues, a feature normally seen in nocturnal species. Each video shows one bumblebee released into an arena; the bee searches for food and then exits. By tracking the path we ask whether the bee's orientation can be explained by path integration.

Funda's role

1. Watch all videos and select the best-quality recordings.
2. Run TRex on each accepted video and obtain its path.
3. Compute a fixed set of behavioural metrics for each video.

Status colours used in this document

- Ready — TRex gives this directly, no extra work needed.
- Compute — a simple formula on the TRex output.
- Decide — needs a threshold or definition before we can compute.

A. What we will measure

Metric	What it tells us	Status
General movement metrics		
Trajectory	The full 2D path of the bee — the foundation of every other metric.	**Ready**
Walking speed	How fast the bee is moving (cm/s).	**Ready**
Acceleration	How fast the bee speeds up or slows down (cm/s²).	**Ready**
Path straightness	How direct the path was — 1 = perfectly straight, 0 = closed loop.	**Compute**
Vector stability	How steady the velocity vector was — 1 = constant motion, 0 = random.	**Compute**
Heading consistency	How tightly the heading clusters around a mean direction.	**Compute**
Pauses (number & duration)	When and how long the bee stopped.	**Decide**
Search events	Bouts of slow movement combined with high turning.	**Decide**

Metric	What it tells us	Status
Drift	How far sideways the bee strayed from a reference line.	**Decide**
Exit angle	Direction the bee crosses any defined boundary.	**Decide**
Orientation recovery	Time to return to baseline heading after a disturbance.	**Decide**
Exploratory vs goal-directed	Share of the trial spent searching vs heading purposefully.	**Decide**

B. What we still need to decide

Topic	What we need to decide
1. Experimental design	
Outbound or inbound?	Do we analyse the trip toward the food, the trip back, or both?
Training vs test trials	Are training and test videos analysed the same way, or differently?
Bee identification	Are bees individually marked? How many trials per bee — do we need repeated-measures statistics?
Conditions & sample size	Full list of conditions (light levels, ocelli-covered, controls) and number of bees per condition?
Match Marie's dataset	Should we compute the same metrics as Marie's analysis so results are directly comparable?
2. Arena, calibration & stimulus	
Arena geometry	Outer / inner / intermediate circle radii (cm)? Centre coordinates? Food location?
Calibration	Real arena width (cm) and camera frame rate (fps)?
Stimulus timing	Is the polarised stimulus on continuously, or only at certain moments?
Stimulus position	Where is the stimulus — overhead, side? This sets the reference axis.
Reference axis (0°)	Is 0° the expected nest direction, arena north, or the stimulus axis?
3. Time events to mark in each video	
Release frame	How do we identify the release moment in each video?
Food-found frame	What counts as 'the bee found the food'? (Lands on it / stops near it / specific behaviour)
Initial heading window	How long after release do we measure the first direction? (1, 3, or 5 s)
Multiple search loops	If the bee leaves and re-enters the inner circle, do we analyse each loop separately or pool them?

Topic	What we need to decide
4. Behaviour thresholds	
Pause rule	Speed below ___ cm/s for at least ___ seconds?
Search rule	Speed below ___ cm/s AND turning above ___ rad/s?
Goal-directed rule	Sliding-window straightness above ___ AND speed above ___ cm/s?
Trial exclusion	Beyond video quality: when do we exclude a trial? (Bee never engages / never exits / partial only)

C. Video quality checklist

☐ **The bee is fully visible from start to end.**

☐ **Only one bee is in the arena.**

☐ **Lighting is stable.**

☐ **Camera is fixed — no shake or drift.**

☐ **Polariser and stimulus log show no equipment issues.**

☐ **The bee performed the full task: release → food → exit.**

☐ **No reflections, dirt, or motion blur block the bee.**

☐ **Trial duration is sufficient to cover all phases.**

☐ **Frame rate is consistent — no dropped frames.**

☐ **Calibration data exists for that recording session.**

D. Standardised pipeline (12 steps)

4. Apply the quality checklist.
5. Calibrate: set the arena width in centimetres.
6. Run TRex (background subtraction, one individual).
7. Inspect the CSV for tracking gaps.
8. Compute the path-integration metrics (group A).
9. Compute the general movement metrics (group B).
10. Add a per-trial summary row to the master CSV.
11. Generate a trajectory plot with the circles overlaid.
12. Back up the CSVs and plots to lab storage.

E. TRex command (single bumblebee)

```
trex \

  -gpu_torch_device cpu \

  -i /full/path/<video>.mp4 \

  -detect_type background_subtraction \

  -track_max_individuals 1 \

  -meta_real_width <ARENA_WIDTH_CM> \

  -track_max_speed 50 \

  -output_format csv \

  -auto_quit true \

  -nowindow true
```